

A Two-Point Group Frame Whose Generator Set Is Disconnected

arXiv automatic math run `fa_banach_001`

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Status

This packet gives a candidate counterexample to Problem 2.19 of Krishna [1]. The problem asks whether the set $\mathcal{F}_G(\pi, \mathcal{X}) \subseteq \mathcal{X}^* \times \mathcal{X}$ of group-p-USF functional-vectors is path connected in the norm topology. The answer is no already for $p = 1$, the two-element group, and the left regular representation on $\ell_1^2(\mathbb{C})$.

Source Question

The source asks the following at the end of Section 2.

in [16]). Later Kaftal, Larson and Zhang noticed that the set of all frame vectors is path-connected in norm topology (see Theorem 8.1 in [20]). Based on these, we raise the following questions.

Problem 2.18. *Given an invertible isometric representation $\pi : G \rightarrow \mathbb{I}^*(\mathcal{X})$ which admits a functional-vector (f, τ) , characterize $\mathcal{F}_G(\pi, \mathcal{X})$. In particular, classify Banach spaces, groups and representations such that*

- (i) $f[\mathbb{I}^*(\pi(G)')^{-1}, \mathbb{I}^*(\pi(G)')]\tau = \mathcal{F}_G(\pi, \mathcal{X})$.
- (ii) $f[\mathbb{I}^*(\pi(G)'')^{-1}, \mathbb{I}^*(\pi(G)'')]\tau = \mathcal{F}_G(\pi, \mathcal{X})$.

Problem 2.19. *Is $\mathcal{F}_G(\pi, \mathcal{X}) \subseteq \mathcal{X}^* \times \mathcal{X}$ path connected in the norm-topology?*

We address Problem 2.19. The same calculation also gives a complete description of $\mathcal{F}_G(\pi, \mathcal{X})$ for this particular representation, which is a small instance of Problem 2.18.

Definitions Used

Let G be a discrete group and let $\pi : G \rightarrow \mathbb{I}^*(\mathcal{X})$ be an invertible isometric representation. Given $f \in \mathcal{X}^*$ and $\tau \in \mathcal{X}$, the associated group-generated functionals and vectors are

$$f_g = f\pi_{g^{-1}}, \quad \tau_g = \pi_g\tau.$$

The pair (f, τ) belongs to $\mathcal{F}_G(\pi, \mathcal{X})$ when $(\{f_g\}_{g \in G}, \{\tau_g\}_{g \in G})$ is a group-p-USF. For $p = 1$ and finite G , this means that the analysis map

$$\theta_f x = (f_g(x))_{g \in G} : \mathcal{X} \rightarrow \ell_1(G)$$

is an isometry and the synthesis map

$$\theta_\tau(a_g)_{g \in G} = \sum_{g \in G} a_g \tau_g$$

is bounded and satisfies the unconditional Schauder reconstruction $\theta_\tau \theta_f = I_{\mathcal{X}}$.

The Counterexample

Let $G = C_2 = \{e, s\}$, with $s^2 = e$. Put $\mathcal{X} = \ell_1(G) = \ell_1^2(\mathbb{C})$, with standard basis δ_e, δ_s and coordinate functionals ζ_e, ζ_s . Let $\pi = \lambda$ be the left regular representation:

$$\lambda_s \delta_e = \delta_s, \quad \lambda_s \delta_s = \delta_e.$$

Then λ is an invertible isometric representation on $\ell_1^2(\mathbb{C})$. The pair (ζ_e, δ_e) is a group-1-USF functional-vector because its analysis and synthesis maps are both the identity on ℓ_1^2 . Hence $\mathcal{F}_G(\lambda, \ell_1^2) \neq \emptyset$.

Lemma 1. *With the preceding notation,*

$$\mathcal{F}_G(\lambda, \ell_1^2) = \{(\alpha \zeta_e, \alpha^{-1} \delta_e) : \alpha \in \mathbb{T}\} \cup \{(\alpha \zeta_s, \alpha^{-1} \delta_s) : \alpha \in \mathbb{T}\}.$$

Proof. Take any $(f, \tau) \in \mathcal{F}_G(\lambda, \ell_1^2)$. Write

$$f = a \zeta_e + b \zeta_s, \quad \tau = u \delta_e + v \delta_s.$$

Relative to the ordered basis (δ_e, δ_s) of both the domain and $\ell_1(G)$, the analysis operator is

$$A = \theta_f = \begin{pmatrix} a & b \\ b & a \end{pmatrix},$$

because the second coordinate functional is $f_s = f \lambda_s = b \zeta_e + a \zeta_s$. The group-1-USF condition says that $A : \ell_1^2 \rightarrow \ell_1^2$ is an isometry. Since the spaces have the same finite dimension, A is a surjective linear isometry.

Surjective linear isometries of complex ℓ_1^2 are monomial. Indeed, an isometry maps the extreme points of the unit ball onto extreme points, and the extreme points of the complex ℓ_1^2 -unit ball are precisely $\gamma \delta_e$ and $\gamma \delta_s$, $|\gamma| = 1$. Therefore each column of A has exactly one nonzero coordinate of modulus one, and the two columns have different supports.

The symmetry of A leaves only two possibilities:

$$A = \alpha I \quad (|\alpha| = 1), \quad \text{or} \quad A = \alpha \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (|\alpha| = 1).$$

The synthesis operator determined by $\tau = u \delta_e + v \delta_s$ has matrix

$$B = \theta_\tau = \begin{pmatrix} u & v \\ v & u \end{pmatrix},$$

because its columns are $\tau_e = \tau$ and $\tau_s = \lambda_s \tau$. The USF reconstruction gives $BA = I$. Hence $B = A^{-1}$.

If $A = \alpha I$, then $B = \alpha^{-1} I$, so $(f, \tau) = (\alpha \zeta_e, \alpha^{-1} \delta_e)$. If $A = \alpha S$, where S is the swap matrix, then $B = \alpha^{-1} S$, so $(f, \tau) = (\alpha \zeta_s, \alpha^{-1} \delta_s)$. Conversely, each pair in these two families plainly has $A = \alpha I$ or $A = \alpha S$ and $B = A^{-1}$, so it is a group-1-USF functional-vector. $\square$

Theorem 1. $\mathcal{F}_G(\lambda, \ell_1^2(\mathbb{C})) \subseteq (\ell_1^2)^* \times \ell_1^2$ is not path connected in the norm topology.

Proof. By the lemma, $\mathcal{F}_G(\lambda, \ell_1^2)$ is the union of the two nonempty sets

$$E_e = \{(\alpha\zeta_e, \alpha^{-1}\delta_e) : \alpha \in \mathbb{T}\}, \quad E_s = \{(\alpha\zeta_s, \alpha^{-1}\delta_s) : \alpha \in \mathbb{T}\}.$$

These sets are separated. For any $\alpha, \beta \in \mathbb{T}$,

$$\|\alpha\zeta_e - \beta\zeta_s\|_{(\ell_1^2)^*} = \|(\alpha, -\beta)\|_{\ell_\infty^2} = 1.$$

Thus, in any of the standard equivalent product norms on $(\ell_1^2)^* \times \ell_1^2$, the distance between E_e and E_s is positive. Consequently E_e and E_s are disjoint relatively open and closed subsets of $\mathcal{F}_G(\lambda, \ell_1^2)$. Since both are nonempty, the set $\mathcal{F}_G(\lambda, \ell_1^2)$ is disconnected, and hence cannot be path connected. $\square$

Verification Notes

No infinite-dimensional or convergence issue is hidden in the construction: the group is finite and all sums have two terms. The only structural input is the elementary finite-dimensional classification of the complex ℓ_1^2 -isometries. The proof also shows that, for this representation, the inclusions from Proposition 2.14 and Theorem 2.17 are equalities: the invertible isometries commuting with the swap are exactly the scalar identity and scalar swap operators.

Novelty Check

The local run indexes were searched for arXiv:2305.01499, the title *Group-Frames for Banach Spaces*, group-p-USF, Banach frames, frame, group representation, orbit, Schauder frame, path connected, and $\mathcal{F}_G$ keywords. No exact prior packet for this question was present.

External searches on June 28, 2026 for combinations of arXiv:2305.01499, the source title, path connected, group-p-USF, and $\mathcal{F}_G$ did not locate a published answer. This should be treated as a bounded novelty check.

Human Review Recommendation

Confirm that Problem 2.19 is meant for the fixed p in the surrounding group-p-USF context. Under that reading, choosing $p = 1$ gives a complete negative answer to the general path-connectedness question.

References

- [1] K. Mahesh Krishna, *Group-Frames for Banach Spaces*, arXiv:2305.01499.